

Human Presence Radar (LD2410C)

Hardware Overview

What is it?

The **LD2410C** is a "mmWave Radar" sensor. Unlike simple motion sensors (PIR) that only see *movement*, this radar transmits high-frequency radio waves to detect Presence.

- It contains a 24 GHz radio transmitter and receiver.
- It can detect stationary humans by sensing the tiny movements of your chest while breathing.

Electrical Explanation

- **Technology:** It uses "FMCW" (Frequency Modulated Continuous Wave) radar technology, like what is used in modern cars.
- **Beam Width:** It sees a cone of about 60° . It cannot see through metal, but it can see through plastic and thin wood (so you can hide it inside a 3D printed case!).
- **Power:** It requires 5V and consumes $\sim 80\text{mA}$.

Key Hardware Facts

- **Connection:** UART Serial (RX / TX), e.g. Pin 6 and Pin 7
- **Default Baud Rate:** 256000 (too fast for SoftwareSerial; change to 9600).
- **Student Baud Rate:** 9600 (required for SoftwareSerial).
- **Logic Voltage:** 5V
- **Signal Flow:** Radar Waves $\rightarrow$ Reflection $\rightarrow$ Antenna $\rightarrow$ LD2410C Chip $\rightarrow$ Serial Data (TX) $\rightarrow$ K-Comp (RX).

Software & Programming

kcomp_LD2410.h

This library manages the high-speed binary data stream and simplifies the radar's reporting.

Usage: `#include <kcomp_LD2410.h>`

Function Reference

Function Name	Description & Parameters
<code>bool initLD2410(uint8_t rx, uint8_t tx, long baud)</code>	Setup: Starts the radar connection. ▪ rx / tx: K-Comp pins (e.g., 6 and 7). ▪ baud: Speed (Use 9600 for SoftwareSerial). ▪ Returns: <code>true</code> if sensor is online.
<code>bool updateLD2410()</code>	Process: MUST be called in <code>loop()</code> . Reads incoming data. ▪ Returns: <code>true</code> if new data arrived.
<code>bool isPresenceDetected()</code>	Check: Returns <code>true</code> if <i>anyone</i> is there (Moving OR Sitting).
<code>bool isMoving()</code>	Check: Returns <code>true</code> if walking/running is detected.

<code>bool isStationary()</code>	Check: Returns <code>true</code> if sitting/breathing is detected.
<code>int getDetectionDistance()</code>	Data: Returns distance to the target in <code>cm</code> .

⚠ Pitfalls & Troubleshooting

- "Sensor not found / Check Wiring"
 - **The Cause:** 90% of the time, the TX/RX wires are swapped.
 - **The Fix:** Check that TX goes to K-Comp Pin 6 (RX) and RX goes to K-Comp Pin 7 (TX).
- "My App doesn't connect anymore!"
 - **The Cause:** You changed the baud rate to 9600 for the K-Comp. The *HLKRadarTool* Bluetooth App expects 256,000.
 - **The Fix:** You cannot use both at the same time. If you need the App, use the Restoration Sketch to reset the sensor to 256,000 baud.

Sample Program

This program displays whether there is someone in the room or not and if so, writes information to the Serial Monitor (sitting, moving, distance).

```
#include <kcomp_LD2410.h>

#define RX_PIN 6 // Remember: TX connects to RX,
#define TX_PIN 7 //           RX connects to TX!

void setup() {
    Serial.begin(115200);

    if (!initLD2410(RX_PIN, TX_PIN, 9600)) { // Initialize at 9600 baud
        Serial.println("CRITICAL ERROR: Sensor not found!");
        while(1); // Stop
    }
    Serial.println("Radar System Online.");
}

void loop() {
    // 1. Read Data (Must be called fast!)
    if (updateLD2410()) {
        // 2. Logic - Print distance and state if detected
        if (isPresenceDetected()) {
            Serial.print("Target at: ");
            Serial.print(getDetectionDistance());
            Serial.print(" cm");

            if(isStationary()) Serial.println(" [Sitting]");
            else Serial.println(" [Moving]");
        } else {
            // 3. Print room empty message
            Serial.println("Room Empty.");
        }
    }
}
```

💡 **Try it out:** You can find the full sample program `SimplePresenceDetectionLD2410C` in the K-Comp examples folder!